

Distance Vector Routing

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Introduction

- Distance Vector Routing is a **dynamic routing algorithm** in which each router maintains a table (vector) that stores:
 - Destination network
 - Distance (cost/metric) to that destination
 - Next hop (neighbour router)
- It is based on the **Bellman–Ford algorithm**.
- Used in protocols like:
 - **Routing Information Protocol (RIP)**
 - **Interior Gateway Routing Protocol (IGRP)**

- Each router:
 - Knows distance to **directly connected neighbours**.
 - Periodically shares its routing table with neighbours.
 - Updates its own table using the **Bellman-Ford formula**.

- **Bellman-Ford Formula**

If router **X** wants distance to destination **Y**:

$$D_X(Y) = \min\{C(X, V) + D_V(Y)\}$$

- Where:
 - $D_X(Y)$ =distance from X to Y
 - $C(X, V)$ =cost from X to neighbor V
 - $D_V(Y)$ =distance from neighbor V to Y
- **Router chooses the minimum cost path.**

Working of Distance Vector Routing

1. Routers share information: Each router sends its distance vector to all its neighbors.

2. Routers update information:

- When a router receives a neighbor's table, it saves it.
- If new information changes the shortest path, the router updates its own table.
- Updates also occur if a link goes down.

$$D_X(Y) = \min\{C(X, V) + D_V(Y)\}$$

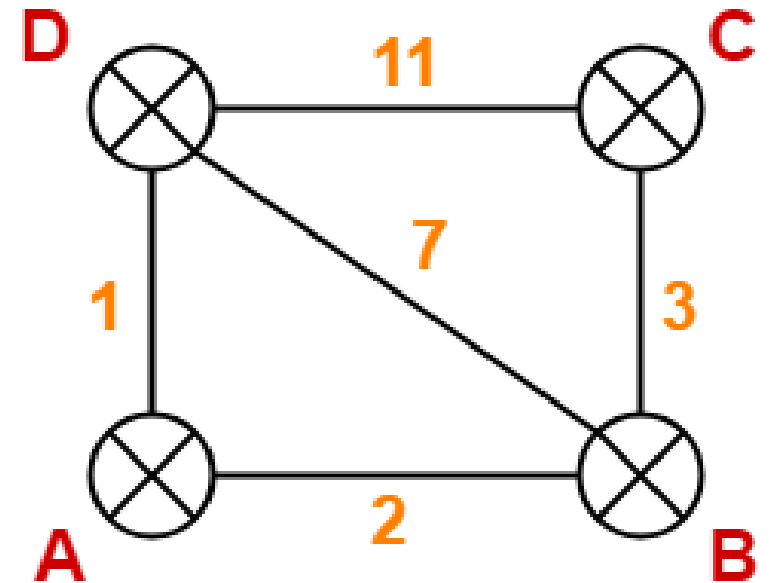
The cost of reaching a destination is calculated using the Bellman-Ford equation:

- Where:
- $D_x(y)$ = cost from router x to destination y
- $C(x,v)$ = cost from router x to neighbor v
- $D_v(y)$ = cost from neighbor v to destination y

Distance Vector Routing Example-

Consider-

- There is a network consisting of 4 routers.
- The weights are mentioned on the edges.
- Weights could be distances or costs or delays.

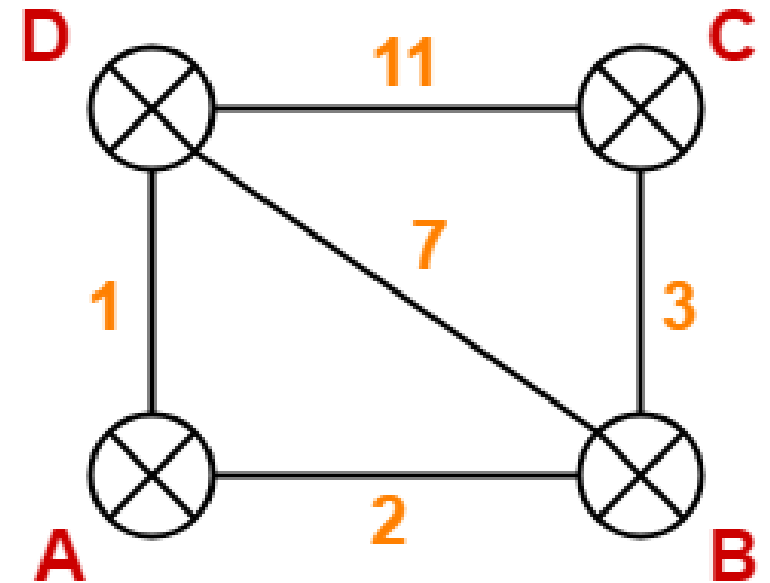


Step-01:

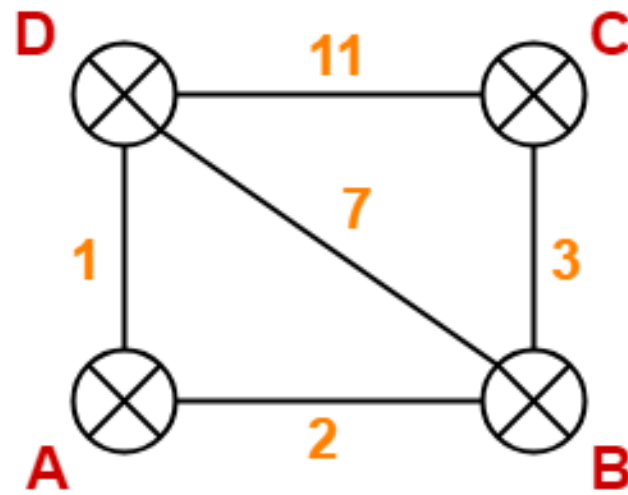
- Each router prepares its routing table using its local knowledge.
- Routing table prepared by each router is shown below-

- At Router A-

Destination	Distance	Next Hop
A	0	A
B	2	B
C	∞	—
D	1	D



Step-01:



At Router B-

Destinati on	Distance	Next Hop
A	2	A
B	0	B
C	3	C
D	7	D

At Router C-

Destinat ion	Distance	Next Hop
A	∞	—
B	3	B
C	0	C
D	11	D

At Router D-

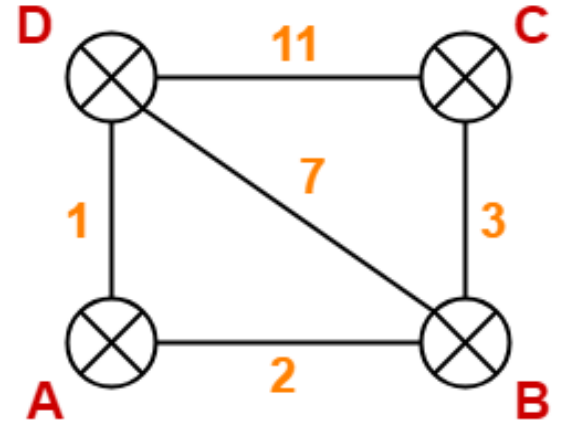
Destinati on	Distance	Next Hop
A	1	A
B	7	B
C	11	C
D	0	D

Step-02:

- Each router exchanges its distance vector obtained in Step-01 with its neighbors.
- After exchanging the distance vectors, each router prepares a new routing table.
-
- This is shown below-

At Router A-

- Router A receives distance vectors from its neighbors B and D.
- Router A prepares a new routing table as-



From B

2
0
3
7

From D

1
7
11
0

Destination	Distance	Next hop
A	0	A
B		
C		
D		

Cost(A→B) = 2 Cost(A→D) = 1

New Routing Table at Router A

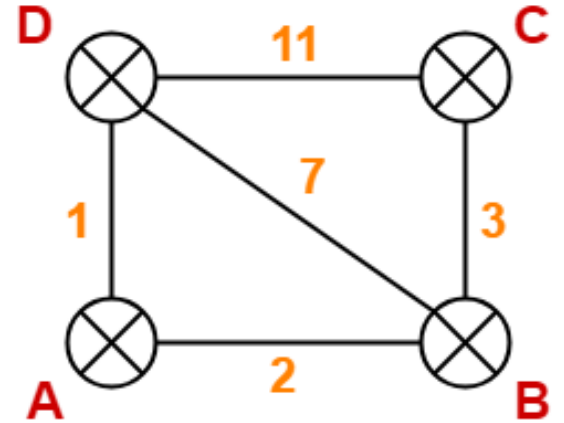
Thus, the new routing table at router A is-

Destination	Distance	Next Hop
A	0	A
B	2	B
C	5	B
D	1	D

- Cost of reaching destination B from router A = $\min \{ 2+0, 1+7 \} = 2$ via B.
- Cost of reaching destination C from router A = $\min \{ 2+3, 1+11 \} = 5$ via B.
- Cost of reaching destination D from router A = $\min \{ 2+7, 1+0 \} = 1$ via D.

At Router B-

- Router B receives distance vectors from its neighbors A, C and D.
- Router B prepares a new routing table as-



Thus, the new routing table at router B is-

From A	From C	From D
0	∞	1
2	3	7
∞	0	11
1	11	0

Cost (B→A) = 2 Cost (B→C) = 3 Cost (B→D) = 7

Destination	Distance	Next hop
A		
B	0	B
C		
D		

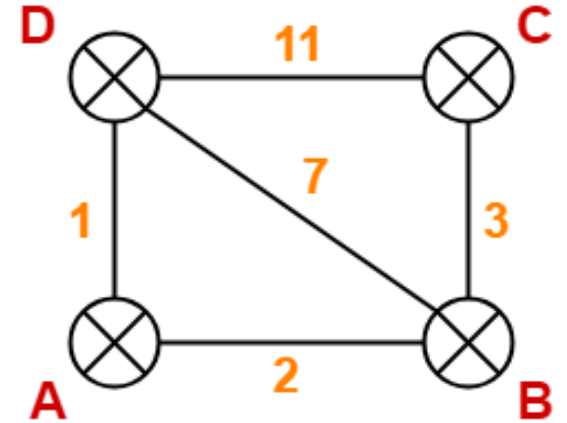
New Routing Table at Router B

Destination	Distance	Next Hop
A	2	A
B	0	B
C	3	C
D	3	A

- Cost of reaching destination A from router B = $\min \{ 2+0, 3+\infty, 7+1 \} = 2$ via A.
- Cost of reaching destination C from router B = $\min \{ 2+\infty, 3+0, 7+11 \} = 3$ via C.
- Cost of reaching destination D from router B = $\min \{ 2+1, 3+11, 7+0 \} = 3$ via A.

- At Router C-

- Router C receives distance vectors from its neighbors B and D.
- Router C prepares a new routing table as-



From B

2
0
3
7

From D

1
7
11
0

Cost (C→B) = 3 Cost (C→D) = 11

Destination	Distance	Next hop
A		
B		
C	0	C
D		

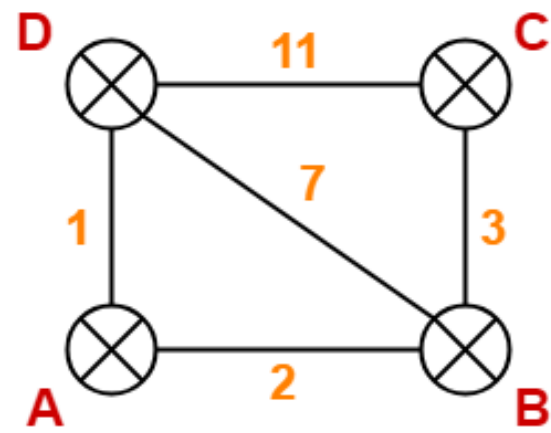
New Routing Table at Router C

Thus, the new routing table at router C is-

Destination	Distance	Next Hop
A	5	B
B	3	B
C	0	C
D	10	B

- Cost of reaching destination A from router C = $\min \{ 3+2, 11+1 \} = 5$ via B.
- Cost of reaching destination B from router C = $\min \{ 3+0, 11+7 \} = 3$ via B.
- Cost of reaching destination D from router C = $\min \{ 3+7, 11+0 \} = 10$ via B.

- At Router D-
- Router D receives distance vectors from its neighbors A, B and C.
- Router D prepares a new routing table as-



From A

0
2
∞
1

From B

2
0
3
7

From C

∞
3
0
11

Cost (D→A) = 1 Cost (D→B) = 7 Cost (D→C) = 11

Destination	Distance	Next hop
A		
B		
C		
D	0	D

New Routing Table at Router D

Thus, the new routing table at router D is-

Destination	Distance	Next Hop
A	1	A
B	3	A
C	10	B
D	0	D

- Cost of reaching destination A from router D = $\min \{ 1+0, 7+2, 11+\infty \} = 1$ via A.
- Cost of reaching destination B from router D = $\min \{ 1+2, 7+0, 11+3 \} = 3$ via A.
- Cost of reaching destination C from router D = $\min \{ 1+\infty, 7+3, 11+0 \} = 10$ via B.

Step-03:

- Each router exchanges its distance vector obtained in Step-02 with its neighboring routers.
- After exchanging the distance vectors, each router prepares a new routing table.
- This is shown below-

At Router A-

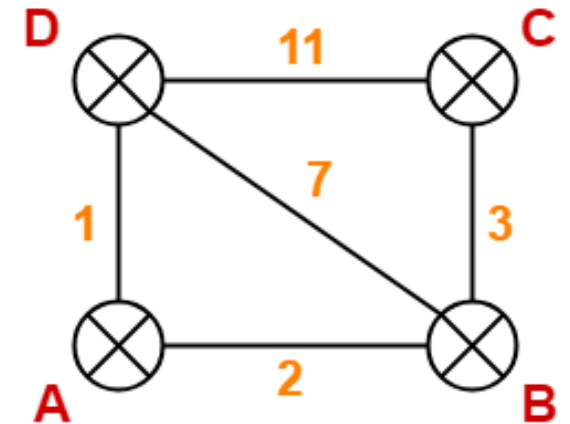
- Router A receives distance vectors from its neighbors B and D.
- Router A prepares a new routing table as-

From B	From D
2	1
0	3
3	10
3	0

Cost(A→B) = 2 Cost(A→D) = 1

Destination	Distance	Next hop
A	0	A
B		
C		
D		

New Routing Table at Router A



Thus, the new routing table at router A is-

Destination	Distance	Next Hop
A	0	A
B	2	B
C	5	B
D	1	D

- Cost of reaching destination B from router A = $\min \{ 2+0, 1+3 \} = 2$ via B.
- Cost of reaching destination C from router A = $\min \{ 2+3, 1+10 \} = 5$ via B.
- Cost of reaching destination D from router A = $\min \{ 2+3, 1+0 \} = 1$ via D.

- At Router B-

- Router B receives distance vectors from its neighbors A, C and D.
- Router B prepares a new routing table as-

From A

0
2
5
1

Cost (B→A) = 2

From C

5
3
0
10

Cost (B→C) = 3

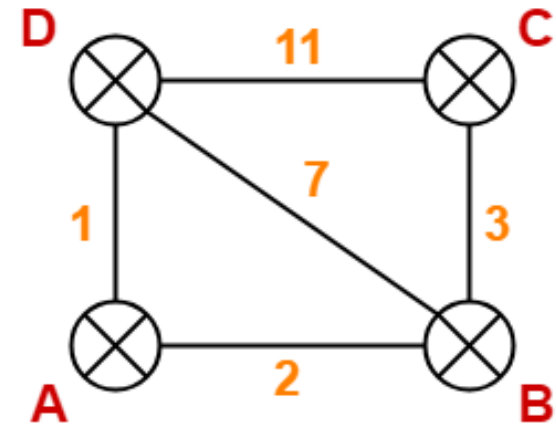
From D

1
3
10
0

Cost (B→D) = 3

Destination	Distance	Next hop
A		
B	0	B
C		
D		

New Routing Table at Router B



Thus, the new routing table at router B is-

Destination	Distance	Next Hop
A	2	A
B	0	B
C	3	C
D	3	A

- Cost of reaching destination A from router B = $\min \{ 2+0, 3+5, 3+1 \} = 2$ via A.
- Cost of reaching destination C from router B = $\min \{ 2+5, 3+0, 3+10 \} = 3$ via C.
- Cost of reaching destination D from router B = $\min \{ 2+1, 3+10, 3+0 \} = 3$ via A.

- At Router C-

- Router C receives distance vectors from its neighbors B and D.
- Router C prepares a new routing table as-

From B

2
0
3
3

Cost (C→B) = 3

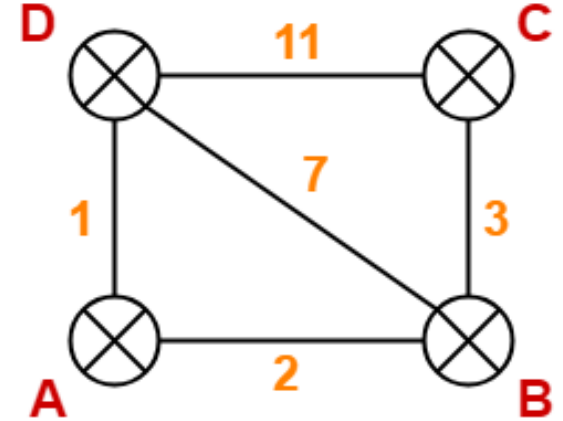
From D

1
3
10
0

Cost (C→D) = 10

Destination	Distance	Next hop
A		
B		
C	0	C
D		

New Routing Table at Router C



Thus, the new routing table at router C is-

Destination	Distance	Next Hop
A	5	B
B	3	B
C	0	C
D	6	B

- At Router D-

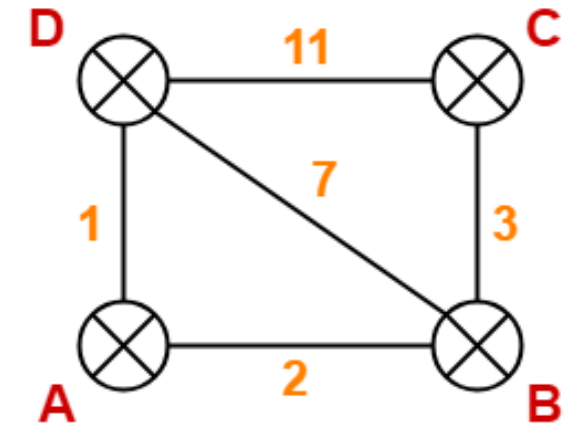
- Router D receives distance vectors from its neighbors A, B and C.
- Router D prepares a new routing table as-

From A	From B	From C	
0	2	5	
2	0	3	
5	3	0	
1	3	10	

Cost (D→A) = 1 Cost (D→B) = 3 Cost (D→C) = 10

Destination	Distance	Next hop
A		
B		
C		
D	0	D

New Routing Table at Router D



Thus, the new routing table at router D is-

Destination	Distance	Next Hop
A	1	A
B	3	A
C	6	A
D	0	D

- Cost of reaching destination A from router D = $\min \{ 1+0, 3+2, 10+5 \} = 1$ via A.
- Cost of reaching destination B from router D = $\min \{ 1+2, 3+0, 10+3 \} = 3$ via A.
- Cost of reaching destination C from router D = $\min \{ 1+5, 3+3, 10+0 \} = 6$ via A.

These will be the final routing tables at each router.

Problems in Distance Vector Routing

1. Routing Loop

- Routers exchange incorrect routing information
- Packets circulate continuously
- Destination is never reached
- Occurs during link failure or slow updates

Why?

- Routers only know neighbour information
- No complete topology knowledge
- Example:
A → B → A → B (looping)

2. Count-to-Infinity Problem

- After link failure, cost increases gradually
- Routers keep increasing metric step-by-step
- Slow detection of unreachable network
- In RIP, infinity = 16 hops
- Example:
2 → 3 → 4 → 5 → ... → 16

3. Slow Convergence

- Network takes time to stabilize
- Temporary wrong routing tables
- Possible loops during update phase

Combined Solutions

Split Horizon

- Router does NOT advertise a route back on the same interface from which it was learned.
- Prevents two routers from misleading each other.
- Reduces routing loops.

Stops incorrect information from going back to its source.

2. Poison Reverse

- If a route fails, router advertises that route with **infinite metric**.
- Explicitly tells neighbors: “This route is unreachable.”
- Speeds up correction process.

Instead of hiding route, it “poisons” it.

3. Triggered Updates

- Router sends update immediately when a change occurs.
- No need to wait for periodic timer.
- Faster convergence.
- **Reduces delay in failure detection**

4. Hold-Down Timer

- After route failure, router ignores suspicious updates for fixed time.
- Prevents accepting unstable or incorrect information.
- Avoids frequent route fluctuations.
- **Gives network time to stabilize.**

Distance Vector problems occur because routers have limited knowledge and rely on neighbour updates.

Solutions work by restricting wrong information and speeding up correct updates.

PRACTICE QUESTION

Consider the following network with the link costs given as follows:

$$A-B = 2$$

$$B-C = 3$$

$$B-D = 4$$

$$C-D = 5$$

Using the Distance Vector Routing Algorithm, answer the following:

- a) Write the initial routing table of Router A.
- b) After one iteration of routing table exchange, update the routing table of Router A.
- c) After convergence, determine the shortest path and total cost from A to C.
- d) After convergence, determine the shortest path and total cost from A to D.
- e) Show the final routing table of Router A after convergence.